

Combining RAG and Agentic Research for Evidence-Grounded Scientific Question Answering

Zhirui Zhu

Independent Researcher
julie1996@foxmail.com

Abstract

We present ThousandOfWind’s hybrid system for GroundLM 2026 LitTraceQA, combining traditional retrieval-augmented generation (RAG) with code-agent research. Agents design question-specific search scripts, revise them as evidence emerges, and reconcile their findings with RAG candidates. The required-test configuration starts with independent research and accepts subsequent revisions through field-level confidence gates and table-structure checks. Using GPT-5.6 Terra in GitHub Copilot CLI, the system raises the 71-question test composite from 0.3567 to 0.7374 and multiple-choice accuracy from 18/50 to 50/50. Macro table cell accuracy reaches 0.2698, leaving structured assembly as the main bottleneck. Our analysis explains the limits of row-key-preserving repair and motivates source-supported structural editing.

1 Introduction

Scientific answers depend on the connection between a fact and its context: a value belongs to a particular experiment, and a table row must join compatible facts across sources. **Shared Task 2: LitTraceQA** at GroundLM 2026 evaluates this connection through paper retrieval, evidence localization, and free-form, multiple-choice (MC), or table answers (GroundLM Organizers, 2026; Liu et al., 2026). It provides 55 development examples; the required test contains 50 MC and 21 table questions.

Our *hybrid system* combines a traditional RAG pipeline (Lewis et al., 2020) with agents that design and adapt a search procedure for each question. The pipeline supplies candidate predictions and serves as the standalone baseline. For the required test, agents first investigate independently, then reconcile these candidates against source evidence. A coordinator accepts revisions separately for papers, evidence, and answers. This workflow

brings together tool-based reasoning (Yao et al., 2023) and source-based research and revision (Gao et al., 2023).

We contribute a task-specific research protocol, an implemented field-selection harness, and an analysis of how research, reconciliation, and repair affect the submitted outputs. Agent research supplies most answers retained through reconciliation; RAG provides additional source candidates. The remaining table weakness exposes a further requirement: preserving the alignment between values, conditions, and rows through answer construction and revision.

2 Combining RAG with Agentic Research

We describe the required-test configuration (Figure 1); development refinements are documented in Section 3.1. The *protocol* specifies research requirements, the *agent* constructs the investigation procedure, and the *harness* coordinates work and assembles predictions.

2.1 Research Protocol: Four Contracts

The research skill comprises a SKILL.md workflow and a completion-policy reference. Together they specify four instructional contracts, reused in source-review prompts.

Target specification. Extract the requested entities, quantities, conditions, units, and output schema. Identify table row keys and the claim made by each MC option.

Search and inspection. Design searches around these constraints, inspect primary PDFs or layout-preserving text, and use contradictory findings to revise the investigation. Exact, alias, and embedding searches provide complementary routes to candidate sources.

Answer and provenance. Associate answer values with their papers, physical pages, source ob-

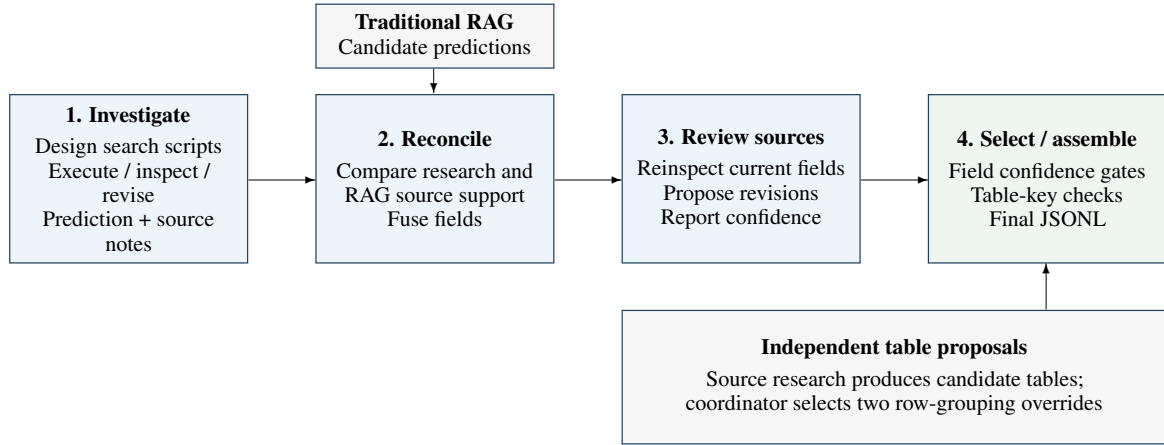


Figure 1: Hybrid RAG and agentic research workflow. Arrows carry predictions or revision proposals: RAG candidates enter at reconciliation, and two selected table overrides enter at assembly. The figure shows the configuration used for the 71-question main test.

jects, conditions, and units. Audit MC options against source facts; join table values under the requested row keys. Return the smallest sufficient supporting paper and evidence sets.

Completion and revision. Complete every requested answer and record uncertain interpretations. The fallback policy orders exact source matches, intended settings, numeric proximity, constraint matching, and a traditional tie-break. Under-specified tables use primary/default source settings with explicit assumptions. Revision proposals require source-based reasons.

2.2 Adaptive Search and Candidate Reconciliation

Each agent owns paper discovery, evidence localization, and answering for its assigned questions. It designs question-specific search and inspection scripts, executes them, reads the results, and adjusts the procedure as new evidence emerges. For example, an unsuccessful phrase search can lead to alias expansion; conflicting values can trigger a script comparing candidate rows under the requested setting. The agent can change search terms, source scope, inspected objects, and comparison logic within the same investigation. Our traditional RAG baseline follows a predefined retrieval and extraction route; the agent constructs this route during the task.

The coordinator supplies traditional predictions only after the agent’s initial predictions are complete. These contain paper IDs, evidence locators, and answers. In the same conversation, the agent

compares the two candidates against the sources and decides separately which papers, locators, and answers to retain or revise. For example, it can retain its answer while adopting a RAG-proposed paper, or remove a redundant locator while keeping the paper set. The coordinator assembles the reconciled outputs and requests further source checks before final field selection.

Example: answering one table field. One test question requests the equation number of a two-dimensional rotation matrix from RomanTex, a texture-synthesis paper in the source corpus (Feng et al., 2025). Figure 2 follows this field from the question to the source and submitted answer. The source shows the matrix symbol R_m and equation label (4); the requested field determines which of these to return.

2.3 Harness: Selecting Field and Table Revisions

Source-review agents inspect the current prediction and return a proposed prediction, field confidence, reasons, and source notes. The assembly script requires complete query coverage and unique proposals. It selects paper, evidence, and answer fields independently, using the minimum available confidence for each field; fields with no recorded confidence receive 0.0000. Thresholds are 0.9900 for papers, evidence, and tables, and 0.9950 for MC. Confidence is an uncalibrated agent judgment.

Table acceptance. A row key is the schema-defined identity of a row, such as a (method,

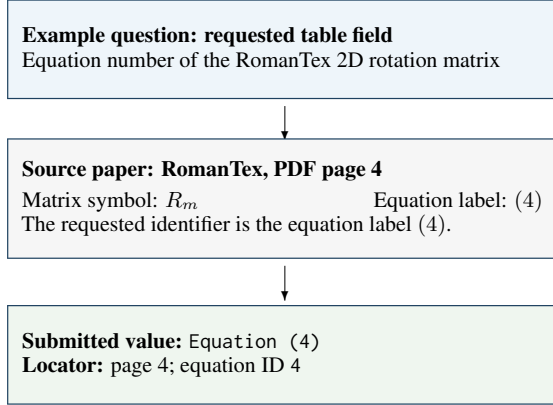


Figure 2: Example from a test-set table question: extracting an equation number from the source paper RomanTex. The three boxes show the requested field, the relevant source content, and the submitted cell with its evidence locator.

dataset) pair; other columns hold the requested values. The conservative table rule has three steps:

1. Extract the ordered row-key tuples from the current and proposed tables. Require unique keys in the current table and exactly the same key sequence in the proposal.
2. Require the available answer-confidence scores to have a minimum of at least 0.9900.
3. Copy the entire proposed table answer when both conditions hold; otherwise retain the current table.

Thus, changing a value under an existing key is eligible; adding, deleting, reordering, or renaming rows requires a structural override. Acceptance operates on the whole table, so all proposed value changes enter together. The gate checks confidence and key structure, leaving each value’s source interpretation to the agent.

Structural overrides. An independent source-reading pass supplies additional table proposals. The final submission incorporates two coordinator-selected whole-table replacements that group multiple quantities under shared row keys. These case-specific overrides extend the conservative rule for the selected tables; routine acceptance follows the three steps above. This division protects existing structure but restricts routine repair to non-key values, a tradeoff examined in Section 3.5.

3 Experimental Setup and Results

3.1 Systems and Evaluation

Traditional pipeline. The baseline combines metadata/full-text retrieval, BGE-M3 embeddings (Chen et al., 2024), reciprocal-rank fusion, and PDF-query localization. It selects three papers and five chunks from a 100-paper reranking pool, with embedding weight 0.2000. Ollama serves qwen3:latest for answer/table synthesis and qwen3-vl:latest for guarded figure interpretation (Yang et al., 2025). MC uses exact, numeric, and token-overlap heuristics, invoking model synthesis when the initial prediction is the first option. Question-pattern handlers route evidence and reconstruct tables; the test baseline includes relational-table postprocessing.

Hybrid execution. GitHub Copilot CLI provides the code-agent runtime and source-access tools. Workers use gpt-5.6-terra: high effort for research and fusion, including development, and xhigh for test-time source review and independent table reconstruction. The agents use released questions, metadata, source PDFs, and extracted-text caches, with their pretrained parameters unchanged. The coordinator manages follow-ups with variable investigation budgets.

Development and test. Both methods were evaluated on development first. The traditional comparison uses matched retrieval/model settings and a table-adapter replay that preserves development outputs and reproduces the test predictions. Hybrid development combines fused paper/evidence fields, agent MC/table answers, and traditional free-form answers. Aggregate development scores selected this answer-type routing. Its answer conversations had prior RAG-candidate exposure, while fresh workers reconstructed evidence. The test configuration adds initial candidate isolation, confidence-gated review, and table safeguards. The coordinator subsequently used aggregate test feedback to select candidates and repair policies.

Supplementary configuration. Table 1 includes the optional 4,901-question test-extra run as a separate reference. It uses GPT-5.6 Sol and Terra at xhigh effort with different completion and repair policies. Its official response reports paper and answer metrics only.

Interpretation and reproducibility. We compare selected system configurations and their

System	Paper	Evid.	MC	Free-form	Row	Cell macro	Cell micro	Score
<i>Required test: 71 questions</i>								
Traditional pipeline	.7048	.0939	.3600	–	.2913	.1627	.2069	.3567
Hybrid system	.9869	.6620	1.0000	–	.4198	.2698	.2759	.7374
<i>Public development: 55 questions</i>								
Traditional pipeline	.7198	.5507	.9024	.8462	.7907	.7225	.7778	–
Hybrid system	.7484	.5618	.9268	.8462	.9293	.8838	.9259	–
<i>Optional test-extra: 4,901 questions (reference configuration)</i>								
Hybrid system	.7390	–	.5215	–	.3216	.1001	.1161	.5267

Table 1: Required-test and development comparisons, with optional test-extra results for reference. Paper, Evid., and Row are F1; MC is accuracy, Free-form is exact match, and Cell columns give macro/micro accuracy. Score is the official test composite; – denotes an unreported metric. Each split uses the configuration described in Section 3.1; test-extra changes worker models and repair policies.

archived intermediate outputs. The Qwen-to-GPT comparison changes models and research budgets together; reconciliation adds both RAG candidates and further source inspection. These comparisons establish system and stage outcomes. Isolating individual mechanisms requires matched-model, matched-budget controls, including a second agent investigation with RAG predictions withheld. Released predictions, score records, model roles, and protocol summaries support artifact inspection;¹ full replay additionally requires complete runtime, sampling, coordinator, and source-version records.

Metrics. Paper and evidence F1 are query-macro averages. Evidence matching uses paper ID, source type, physical page, and table/figure identifiers. Span text and row/column coordinates are excluded from this key; textual entailment and cell-level support require separate assessment. Table row F1 compares row keys; cell accuracy scores non-key fields after row alignment, with macro averaging across table questions and micro pooling scored cells. We report the official composite separately for each test split.

3.2 Core-Test Results

Table 1 shows a composite gain of 0.3807 and MC improvement from 18/50 to 50/50. Paper F1 reaches 0.9869 and Evidence F1 0.6620. Table performance improves more modestly: row F1 reaches 0.4198 and macro cell accuracy 0.2698. This contrast separates successful source discovery and option selection from the remaining difficulty of constructing tables.

We examine this result in three steps: where the fixed pipeline loses accuracy, what research and

reconciliation each contribute, and why subsequent repair leaves table errors.

3.3 Where the Fixed Pipeline Loses Accuracy

With matching retrieval settings and Qwen answer-model digests, traditional Paper F1 changes from 0.7198 on development to 0.7048 on test, while Evidence F1 falls from 0.5507 to 0.0939 and MC accuracy from 0.9024 to 0.3600. The main loss occurs after paper retrieval, in locating the required evidence and selecting an answer.

Two implementation choices help explain this vulnerability. Question-pattern routing commits to source types and extraction handlers early, so unfamiliar wording or layout can send a relevant paper through an unsuitable handler. The synthesis gate revisits only first-option heuristic predictions; errors assigned to other options survive that gate. Repeated tuning on 55 examples can embed familiar patterns in these fixed decisions, a form of *procedural overfitting*. Source accessibility and question-distribution changes remain additional explanations for the cross-split gap. The agent offers a concrete recovery mechanism: after inspecting an unhelpful result, it can change the search terms, source object, or extraction procedure within the same question.

3.4 RAG and Agent Contributions

Agent research supplies the answer foundation.

Before receiving RAG predictions, agents already answer 48/50 required-test MC questions correctly. Reconciliation retains 49/50 of these labels and 20/21 table answers, with MC and table scores unchanged. Most reconciled answers therefore come directly from the initial investigation. The final source-review stage subsequently brings MC accuracy to 50/50.

¹<https://github.com/ThousandOfWind/LitTraceQA>

RAG contributes alternative candidates. Both components retrieve papers, locate evidence, and generate answers; their outputs can agree or reveal different source choices. In the archived reconciliation step, the sole paper-set replacement adopts a RAG candidate absent from the agent’s initial output. Across this step, Paper F1 rises from 0.9728 to 0.9869 and Evidence F1 from 0.5657 to 0.6150. The observed benefit is better source attribution while most answers are retained. This gives RAG a concrete complementary role even though its standalone answer accuracy is lower.

The useful division of work is thus task-dependent. On required-test MC and tables, agent answers form the basis for reconciliation; on development free-form questions, the selected hybrid retains RAG answers. Field-level reconciliation accommodates both patterns by selecting source-supported content separately for each output field.

3.5 Table Errors and Repair Tradeoffs

The answer foundation is much weaker for tables. Development macro cell accuracy rises from 0.7225 to 0.8838, with 25/27 scored cells correct, yet required-test accuracy reaches only 0.2698. The small development cell sample covers a limited range of row identities and source conditions. Each test table requires binding a value to the correct method, setting, unit, and row; finding the right paper establishes only the source of those facts. Our agents make these binding decisions through source reading, while the harness checks confidence and row-key structure. Uneven cell-level provenance leaves much of this alignment dependent on agent judgment.

The conservative rule in Section 2.3 protects existing keys but freezes their mistakes. A missing row stays missing, a mislabeled key stays mislabeled, and duplicate keys block ordinary table updates. Selective repair leaves row F1 at 0.4198; the two explicit grouping overrides also leave aggregate scores unchanged. This is a restriction on which errors the repair can reach: greater confidence in a proposed value still leaves an incorrect row identity untouched.

Rejected full-replacement diagnostic. To expand this repair space, we evaluated replacing evidence sets and entire tables after independent source research. Rebuilding a table allows missing rows, mislabeled keys, and duplicate rows to be reconsidered together; renewed evidence search can

also recover support for the revised answer. The diagnostic adopts rebuilt paper sets, replaces evidence sets at confidence 0.9900, and accepts rebuilt tables unconditionally. Compared with selective repair, row F1 rises from 0.4198 to 0.4285, but Evidence F1 falls from 0.6620 to 0.5033 and macro cell accuracy from 0.2698 to 0.2302; Paper F1 and MC stay unchanged. We rejected this replacement policy and excluded it from the final submission because the small row gain accompanies substantial evidence and cell regressions. Whole-output acceptance couples structural corrections to every other proposed content change. The diagnostic establishes the regression of this replacement policy as a whole; its aggregate feedback leaves the individual error causes unresolved.

The repair objective should therefore be *comparative*: establish why a proposed edit improves the current field and preserves its supported content. Candidate confidence estimates plausibility; beneficial replacement additionally requires comparison with the retained answer. A promising next design stores source-backed value-condition tuples and applies explicit row and cell edits, checking both the defect being repaired and the supported content being retained.

4 Conclusion

Combining RAG candidates with task-adaptive agent research raises the LitTraceQA test composite from 0.3567 to 0.7374. The stage analysis clarifies the method’s division of work: agent research supplies most reconciled answers, RAG adds alternative source candidates, and field-level selection governs subsequent revisions. The main unresolved problem is carrying source facts into correct table relations. Ordinary repair preserves row identities, leaving structural errors outside its reach. Source-supported row and cell edits provide a concrete next direction for extending that repair space while retaining correct content.

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